

A Survey of Singing Voice Synthesis



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Abstract Singing voice synthesis (SVS) is a task of synthesizing singing audio according to music score and lyrics. SVS is a sub-research direction of speech synthesis. Singing voice is more complex and changeable than speaking voice in pitch, duration, and energy, so modeling singing voice is more difficult. With the rapid development of deep learning, more and more researchers in academia and industry have devoted themselves to research, and the research goal is to generate natural, expressive and human-like singing voices. This paper analyzes and compares the current challenges and progress in the research of SVS from three aspects: datasets, data augmentation strategies, and SVS models. Among them, the key point is to compare how the two-stage and fully end-to-end SVS models solve various problems in this field. Finally, we summarize the current research situation and discuss future research directions. It is expected to bring some inspiration to researchers in this field.

Keywords Singing voice synthesis · Acoustic model · Vocoder

1 Introduction

At the end of the 1950s, the earliest computer music project studied by Bell Labs realized the singing voice synthesis (SVS), and a representative physical acoustic model is Kelly and Lochbaum [1]. After the development of traditional methods such as unit splicing and statistical parameters, the current mainstream method is based on deep learning. With the rise of concepts such as the metaverse and virtual singer, SVS has attracted more attention from researchers in academia and industry. Although SVS has been developed for many years, there is no formal definition.

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The general definition of SVS in the research is to generate singing voices based on music score and lyrics. Li et al. [2] proposed that SVS refers to the synthesis of simulated singing voice through computer algorithms, which can be seen as an extension of speech synthesis technology and a comprehensive application of music synthesis technology. Singing is more exaggerated in pitch, duration, and energy than speaking, and the singing voice is more expressive.

Although there have been some reviews about SVS [3, 4], they mainly summarize and compare from the perspective of SVS models. We think that the above work is not enough. In the SVS research based on deep learning, data and model are equally important, so data should also be summarized. This paper mainly focuses on the datasets, data augmentation strategies, two-stage SVS models, and fully end-to-end SVS models. The two-stage SVS models and fully end-to-end SVS models will be introduced from the modeling method.

2 Research About Datasets

The production of SVS datasets requires high-quality recordings of pure singer voice, the phoneme tagging of lyrics, and the alignment of singing audio with lyrics. Among them, the most difficult part is the alignment between lyrics text and audio, and it needs to be accurate on a small time scale. However, high-quality datasets still require careful manual inspection. Table 1 shows the common singing voice synthesis public datasets.

Table 1 Singing voice synthesis public datasets

Datasets	Year	Language	Total duration	Singers
NUS-48E [5]	2013	English	2.8 h	Multiple
VocalSet [6]	2018	English	10.1 h	Multiple
Jingju [7]	2018	Chinese(Dialect)	~2 h	Multiple
CSD [8]	2020	English/Korean	2.34 h/2.52 h	Single
JVS-MuSiC [9]	2020	Japanese	~2 h	Multiple
NIT-SONG070-F001 [10]	2020	Japanese	1.2 h	Single
MPop600 [11]	2021	Chinese (Mandarin)	10 h	Multiple
OpenSinger [12]	2021	Chinese (Mandarin)	50 h	Multiple
Kiritan [13]	2021	Japanese	3.5 h	Single
Opencpop [14]	2022	Chinese (Mandarin)	5.2 h	Single
PopCS [15]	2022	Chinese (Mandarin)	5.89 h	Single
M4Singer [16]	2022	Chinese (Mandarin)	30.59 h	multiple
Ofuton [17]	2022	Japanese	~1 h	Single
Namine Ritsu [18]	2022	Japanese	4.35 h	Single

We can find that the languages of the current public datasets are mainly Chinese, English, Japanese, and Korean. In the past two years, many Chinese datasets with long duration and high quality have been released. In addition to the above public datasets, some studies [19–22] also use the non-free singing voice datasets provided by Databaker technology company or some internal non-public datasets.

3 Data Augmentation Strategies

Before large-scale high-quality datasets are available, the results of the model on public datasets with short duration need to be improved. So some researchers begin to study data augmentation strategies to expand the duration and diversity of training data. The simple way of data augmentation strategy is to expand data offline according to certain rules. Offline data augmentation is separated from model training. However, the complex way is to expand data online according to certain strategies during the training process. Meanwhile, the data augmentation and training model are integrated.

Zhao et al. [23] proposed a simple data enhancement technology for the Chinese SVS task. They expand the data from time domain and frequency domain. In the time domain, they use five methods to enhance audio data: clipping, inverting, flipping, noising, and time extension. These methods can also be used in combination. In the frequency domain, they scale the pitch of data to enhance data diversity. After that, they add the expanded data to the original data and then use this data to train the model. Zhang et al. [24] proposed a new multi-singer pre-training data augmentation, which improves robustness and naturalness of SVS. In order to avoid the impact of different input audio sample lengths, they divide each song into three small segments with different time intervals. They explore whether the SVS model based on Transformer [25] is affected by different lengths of the input sequence. In addition, Guo et al. [26] proposed Pitch Augmentation (PA) and Mix-up Augmentation (MA) data augmentation strategies. PA is similar to the method in [23]. They multiply or divide the F0 sequence of the audio sample by $\sqrt[12]{2}$, maintaining the aperiodic spectral envelope and harmonic spectral envelope unchanged. Then they send them into the WORLD vocoder [27] to obtain the singing audio. MA refers to selecting two samples i and j randomly from each mini-batch based on an encoder-decoder structure acoustic model. They mix the encoder output implicit representation H_i and H_j by a weight λ , thus the mixed implicit representation H_{mix} can be written as Eq. (1).

$$H_{mix} = \lambda * H_i + (1 - \lambda) * H_j \quad (1)$$

Last, H_{mix} is input to the decoder to obtain the predicted acoustic feature $\hat{Y}_{mix}$. The loss function L_{mix} of MA is illustrated in Eq. (2). They finally add the loss L_{mix} to the original L1 loss to train the acoustic model. Unlike [23, 24, 26] Mix up samples in the model training process and do not need to augment the data before training.

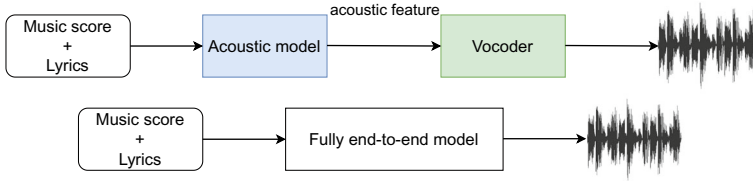


Fig. 1 The pipeline comparison between two-stage model and fully end-to-end model

$$L_{mix} = \lambda * |\hat{Y}_{mix} - Y_i| + (1 - \lambda) * |\hat{Y}_{mix} - Y_j| \quad (2)$$

where Y_i and Y_j are the real acoustic feature of samples i and j .

4 Research About SVS Models

At present, the research of the SVS model based on deep learning can be mainly divided into two categories: two-stage models and fully end-to-end models. Figure 1 shows the pipeline comparison between the two-stage model and the full end-to-end model. The two-stage model contains an acoustic model and a vocoder which are trained separately. The input of acoustic model are music score and the grapheme/phoneme corresponding to the lyrics and the output is the acoustic feature (e.g., Mel-spectrum or Linear-spectrum). The vocoder takes in the acoustic feature and then outputs the synthesized audio waveform. In the study of acoustic models, researchers focus on how to make the predicted mel-spectrum close to the mel-spectrum of real singing audio, especially the pitch, duration, energy, and pronunciation of singing. In the research of vocoder, researchers focus on how to improve the quality of synthetic singing voices on the premise of reducing network parameters and accelerating inference speed. Fully end-to-end model combines the above two stages into one stage, i.e., incorporates the music score and the grapheme/phoneme corresponding to the lyrics and generates waveform directly.

4.1 Two-Stage Models

According to the different modeling methods, we introduce two-stage models in the following directions: Sequence to sequence (Seq2seq), Variational auto-encoder (VAE), Generative Adversarial Network (GAN), and diffusion models. At the same time, we will also introduce the challenges that each model solves. Table 2 compares the representative SVS models mentioned in this paper.

Table 2 Singing voice synthesis models (the two-stage model is called TS for short. The fully end-to-end model is called E2E for short. The external vocoder is called Ex-vocoder for short)

Model	TS/E2E	Modeling	Acoustic model/vocoder	Ex-vocoder
ByteSing [28]	TS	Seq2seq	Acoustic model	WaveRNN
Singing-Tacotron [19]	TS	Seq2seq	Acoustic model	LPCNet
FFT-NPSS [29]	TS	Seq2seq	Acoustic model	WORLD
PE Seq2Seq [30]	TS	Seq2seq	Acoustic model	Griffin-Lim
XiaoiceSing [20]	TS	Seq2seq	Acoustic model	WORLD
Deepsinger [31]	TS	Seq2seq	Acoustic model	Griffin-Lim
KaraSinger [22]	TS	VQ-VAE	Acoustic model	MelGAN
CGAN-mgc [32]	TS	GAN	Acoustic model	STRAIGHT
HiFiSinger [21]	TS	GAN	Acoustic model	PWG
N-Singer [33]	TS	GAN	Acoustic model	PWG
WGANSing [34]	TS	GAN	Acoustic model	WORLD
SVS-BEGAN [35]	TS	GAN	Acoustic model	Griffin-Lim
WeSinger2 [36]	TS	GAN	Acoustic model and vocoder	Unneeded
Multi-Singer [12]	TS	GAN	Vocoder	Unneeded
Avocodo [37]	TS	GAN	Vocoder	Unneeded
DiffSinger [15]	TS	Diffusion	Acoustic model	HiFi-GAN
Learn2Sing2.0 [38]	TS	Diffusion	Acoustic model	RefineGAN
VISinger [39]	E2E	VAE + GAN	No	Unneeded
VISinger2 [40]	E2E	VAE + GAN	No	Unneeded

4.1.1 Seq2seq-Based Models

Among the Seq2seq models, ByteSing [28], Singing-Tacotron [19], FFT-NPSS [29], Seq2Seq with perceptual entropy (PE) loss [30] are autoregressive models. Bytesing is a Chinese SVS system that consists of an acoustic model and a WaveRNN [41] vocoder both of which has an encoder-decoder structure. The encoder of ByteSing’s acoustic model is a CBHG encoder as that in Tacotron [42] which is composed of three parts: 1-D convolution bank, highway network, and bidirectional Gate Recurrent Unit (GRU). The decoder of ByteSing’s acoustic model follows the decoder in Tacotron2 [43]. It uses an autoregressive recurrent neural network (RNN) to predict mel-spectrum. The input sequences are expanded by an auxiliary duration prediction model, so the attention module can achieve fast convergence. This acoustic model can improve the controllability, stability, and rhythm prediction accuracy of the model. The structure of the Singing-Tacotron model is similar to that of ByteSing. Singing-Tacotron is composed of an acoustic model and an LPCNet [44] vocoder. Its key innovation lies in the design of the acoustic model which consists of the content encoder, duration encoder, global duration control attention module, and decoder.

The content encoder encodes lyrics and pitches to hidden features while the duration encoder encodes durations and tempo to global transition tokens. The duration of each phoneme is controlled by the global duration attention module according to the global transition tokens. Finally, the decoder generates a mel-spectrum acoustic feature provided by all the information. The advantage of their acoustic model is that they use misaligned singing datasets. In the model training, the duration encoder makes the data alignment more accurate, and the dynamic filter improves the stability of the model training.

Autoregressive model predicts tokens one by one, thus the generation process is slow. To tackle this, some studies focus on non-autoregressive (NAR) model. The parallel method speeds up generation progress in the meantime retains generation quality as autoregressive models have. XiaoiceSing [20] and DeepSinger [31] are two representative NAR models. XiaoiceSing is composed of an acoustic model modified from Fastspeech [45] and a WORLD vocoder. The acoustic model also has an encoder-decoder structure. The encoder converts phoneme ID, note duration, and note pitch into dense vector. Duration predictor obtains the duration of the phoneme from dense vector then a length regulator is used to extend dense vector sequence according to the predicted duration of phoneme. The decoder is designed to generate MGC, BAP, V/UV, and F0 acoustic features according to the extended dense vector. In order to alleviate the out-of-tune problem, the researchers also add a pitch residual connection to the F0 predicted by decoder. DeepSinger uses the singing data collected from the internet. The quality of data is poor. DeepSinger is composed of an acoustic model modified from Fastspeech and uses griffin-lim [46] vocoder to generate waveform. The acoustic model combined with a reference encoder can capture the singer’s timbre from noisy data.

4.1.2 VAE-Based Models

VAE is rarely used in the two-stage models. We only find that KaraSinger employs VQ-VAE. KaraSinger [22] is a SVS model which does not need music score input. The rhythm and melody are automatically generated by the machine. KaraSinger is composed of a hierarchical VQ-VAE with mel-spectrum as input and a language model with lyrics as input. VQ-VAE converts the mel-spectrum of real audio into discrete code sequences to train the language model, which generates new singing voice audio. The whole model has relatively few parameters and can realize fast training and inference.

4.1.3 GAN-Based Models

GAN is a kind of generation network which is widely used as the acoustic model and vocoder in the two-stage model. In terms of acoustic model, Hono et al. [32] proposed two SVS models based on GAN, namely GAN-mgc and CGAN-mgc. The generators of these two models are the same, the inputs of which are linguistic and musical fea-

tures. Considering that the discriminator of GAN-mgc can not capture the difference between the real and generated acoustic features accurately, CGAN-mgc employ a discriminator that takes in more linguistic and musical features. Their method can effectively improve the naturalness of singing. The target of HiFiSinger [21] is high-fidelity SVS. It is composed of an acoustic model modified from FastSpeech and a Parallel WaveGAN (PWG) vocoder. The acoustic model of HiFiSinger is similar to that of XiaoIceSing. It focuses on solving two problems. First, to deal with the larger frequency range caused by the high sampling rate audio, it exploits a sub-frequency GAN to model three sub-spectrums divided from the origin 80-dimensional mel-spectrum. Second, the researchers design a multi-length GAN to solve the long waveform generation problem resulting from high sampling rate. In order to alleviate the problem of inaccurate pronunciation in Korean SVS, N-Singer [33] adds a discriminator to determine the harmonic characteristics of voiced segments and the noise of unvoiced segments. WGANsing [34] is essentially a DCGAN which uses Wasserstein GAN to optimize the generation. The generator of WGANsing has a U-Net-like structure which outputs mel-spectrum. The discriminator compares the generated mel-spectrum with the corresponding ground truth mel-spectrum. In order to solve the problem of discontinuity when generating mel-spectrum, Choi et al. [35] proposed an autoregressive SVS-BEGAN. BEGAN is helpful for the stable training of the model and could improve the quality of generating mel-spectrum. The acoustic model and vocoder of WeSinger2 [36] are two GANs. The generator of acoustic model is based on Transformer, and the discriminator is a conditional multi random region discriminator. This discriminator can effectively avoid over-smoothing in spectral prediction and improve the expression ability of SVS. The vocoder is an improved HiFi-GAN which takes in not only mel-spectrum but also 88 discrete piano keys. Multiple adversarial condition discriminators are used in the waveform domain, while multi-scale distance functions are utilized in the frequency domain to optimize the system.

In terms of vocoder model, there is not much research on neural network vocoders besides traditional non-neural network vocoders for SVS. The neural vocoder based on GAN is widely used because of its fast inference speed, compact network structure, and ability to generate high-quality voice. At present, the widely used vocoders are HiFi-GAN [47], MelGAN [48], Parallel WaveGAN [49], etc. While these vocoders have poor performance when synthesizing the singing voice of unseen singers. To solve this problem, Huang et al. [12] proposed Multi-Singer, a fast multi-singer vocoder based on GAN. Multi-Singer uses singer conditioned discriminator to strengthen the modeling ability of singer identity. They also proposed an auxiliary loss of singer perception to monitor the reconstruction of singer identity in the spectrum envelope. Existing GAN-based vocoders have the problem of focusing on noise artifacts in low-frequency bands, which may reduce the quality of the synthesized waveform. Bak et al. [37] proposed a GAN-structured vocoder called Avocado. They introduce a collaborative multi-band discriminator as well as a sub-band discriminator to model spectrum of different frequency regions better.

4.1.4 Diffusion Model-Based Models

Different from GAN, the training process of diffusion model is stable. Diffusion model has recently shown remarkable performance in image generation tasks. Because the mel-spectrum in SVS can be also regarded as an image, diffusion model has been introduced into the research of SVS. The diffusion process can be divided into two parts: noising and denoising. The process of noising is to add noise that obeys a certain probability distribution to the original image so that the image becomes a random noise image gradually. In denoising phase, noise is removed step by step from a noise point to resume a real sample.

DiffSinger [15] is a SVS acoustic model based on the diffusion probability model. It takes the music score and lyrics as the condition and adds a denoiser in addition to the basic encoder-decoder structure. It also has a shallow diffusion mechanism, which effectively reduces the diffusion steps and accelerates the training process. Diff-WGAN [50] is an acoustic model based on the combination of diffusion denoising probability model and WGAN. The addition of WGAN realizes the purpose of promoting music expression and enhancing high-frequency acoustic detail learning. Learn2Sing2.0 [38] explores the SVS for target speakers without singing training data. It models the singer and style whose input is source speaker ID. After predicting mel-spectrum of source speaker, a diffusion decoder adds target speaker information which is used to predict the mel-spectrum of target speaker.

4.2 Fully End-to-End Models

Although the two-stage SVS models have achieved good results, there are still problems of error accumulation and mismatch between training and inference. In the two-stage model, the acoustic model and vocoder are trained separately. The vocoder uses the acoustic features of the real singing voice data during training and the predicted acoustic features during inference, which ultimately leads to the degradation of the model performance during inference. In order to solve this mismatch problem, the fully end-to-end SVS model has been studied.

Compared with the two-stage model, the acoustic model and vocoder in fully end-to-end model are trained together. The inputs of the model are music score and lyrics, and the output is singing audio waveform. Up to now, there are few end-to-end SVS models among which VISinger [39] and VISinger2 [40] are representative. VISinger is composed of prior encoder, posterior encoder, and decoder and discriminator. The inputs of prior encoder are phoneme ID, note pitch ID, and note duration, and the output is a prior distribution. The input of posterior encoder is a real raw waveform, and the output is hidden representation. Decoder predicts the hidden representation into singing waveform. The discriminator uses the multi-period discriminator and multi-scale discriminator in HiFi-GAN. VISinger2 uses digital signal processing method to solve the problem of meaningless text-to-phase mapping, poor modeling of voiced segments, and low sampling rate in VISinger.

5 Conclusion and Prospect

5.1 Conclusion

This paper analyzes and compares the recent research works about SVS. With the release of high-quality public datasets, we basically don't need to worry about data shortages. If you want to use more data, some studies have proved the effectiveness of some data augmentation methods. From the model aspect, most of the mainstream research is still focused on acoustic models and vocoder two-stage models and researchers pay more attention to the design of acoustic models. Most of the works are to enhance the expressiveness of singing voice by improving the acoustic model. NAR model in acoustic model can achieve almost the same performance as autoregressive model does in terms of generating quality with the advantage of fast generation speed. There are few studies on the full end-to-end model, but the performance of VISinger is comparable to the two-stage models.

5.2 Prospect

5.2.1 Research on Emotional SVS

Singers must blend into their own emotion when singing, and we believe that when the machine synthesizes an emotional singing voice, it should first automatically analyze the emotional categories of lyrics, then synthesize the corresponding emotional singing voice. So it is also a great challenge to use the limited training audio data of the singer to decouple the emotion representation and automatically add emotion when synthesizing.

5.2.2 Research on SVS with Accompaniment Data

At present, the research of SVS mainly focuses on the voice of singers. But this is not consistent with the actual situation. In addition to the singing voice, a song is accompanied by musical instruments. How to use the audio data of songs with accompaniment to train and generate songs with accompaniment has more significance and application value. It can also reduce the cost of building datasets.

5.2.3 Research on Multilingual SVS

The current SVS model is a single language data training model, and there is no research work to train multilingual data together. If the multilingual data are trained together, the phoneme symbols of different language lyrics are similar, which may

cause the situation that the synthesized singing voice can not be sung well in any language. How to construct a new paradigm of multilingual SVS model is a new direction.

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References

1. Cook PR (1996) Singing voice synthesis: history, current work, and future directions. *Comput Music J*
2. Li W, Li Z, Shao X (2019) The blending of audio music and computer—audio music technology. Fudan University Press, Shanghai (in Chinese)
3. Marti U, Jordi B, Masataka G, Tomoyasu N, Sundberg J (2015) Expression control in singing voice synthesis, features, approaches, evaluation, and challenges. *IEEE Signal Process Mag*
4. Cho YP, Yang FR, Chang YC, Cheng CT, Wang XH, Liu YW (2021) A survey on recent deep learning-driven singing voice synthesis systems. In: *IEEE AIVR*, Taiwan
5. Duan Z, Fang H, Li B, Sim KC, Wang Y (2013) The NUS sung and spoken lyrics corpus: a quantitative comparison of singing and speech. In: *APSIPA*, Taiwan
6. Wilkins J, Seetharaman P, Wahl A, Pardo B (2018) Vocalset: a singing voice dataset. In: *ISMIR*, Paris
7. Gong R, Repetto RC, Serra X (2017) Creating an a cappella singing audio dataset for automatic Jingju singing evaluation research. *DLfM*, Shanghai
8. Choi S, Kim W, Park S, Yong S, Nam J (2020) Children's song dataset for singing voice research. *ISMIR*, Montréal
9. Tamaru H, Takamichi S, Tanji N, Saruwatari H (2020) Japanese multispeaker singing-voice corpus. *CoRR*, Jvs-music
10. HTS Group (2016) The Nitech Japanese speech database nit song070 f001. https://hts.sp.nitech.ac.jp/archives/2.3/HTS-demo_NIT-SONG070-F001.tar.bz2
11. Chu C-C, Yang F-R, Lee Y-J, Liu Y-W, Shan-Hung W (2020) Mpop600: a mandarin popular song database with aligned audio, lyrics, and musical scores for singing voice synthesis. *APSIPA*, Auckland
12. Huang R, Chen F, Ren Y, Liu J, Cui C, Zhao Z (2021) Multi-singer: fast multi-singer singing voice vocoder with a large-scale corpus. *ACM Multimedia*, Chengdu
13. Ogawa I, Morise M (2021) Tohoku Kiritan singing database: a singing database for statistical parametric singing synthesis using Japanese pop songs. *Acoust Sci Technol*
14. Wang Y, Wang X, Zhu P, Jie W, Li H, Xue H, Zhang Y, Xie L, Bi M (2022) Opencpop: a high-quality open source Chinese popular song corpus for singing voice synthesis. *INTERSPEECH*, Incheon
15. Liu J, Li C, Ren Y, Chen F, Zhao Z (2022) Diffsinger: singing voice synthesis via shallow diffusion mechanism. *AAAI*, Vancouver
16. Zhang L, Li R, Wang S, Deng L, Liu J, Ren Y, He J, Huang R, Zhu J, Chen X, Zhao Z (2022) M4singer: a multi-style, multi-singer and musical score provided mandarin singing corpus. *NeurIPS* (online)
17. Ofuton (2016) Ofuton database. <https://sites.google.com/view/oftn-utagoedb/%E3%83%9B%E3%83%BC%E3%83%A0>
18. Canon Group (2022) Namine Ritsu's Japanese singing database ver2. <https://drive.google.com/file/d/1Nv3qwZ6DSr80W9qcV9ZBg5urKFtdPp5/view>
19. Wang T, Fu R, Yi J, Wen Z, Tao J (2022) Singing-tacotron: global duration control attention and dynamic filter for end-to-end singing voice synthesis. *DDAM@MM*, Lisbon

20. Lu P, Wu J, Luan J, Tan X, Zhou L (2020) Xiaoice-sing: a high-quality and integrated singing voice synthesis system. INTERSPEECH, Shanghai
21. Xu JC, Luan TJ, Tao Q, Liu T-Y (2020) Towards high-fidelity neural singing voice synthesis. CoRR, Hifisinger
22. Liao C-F, Liu J-Y, Yang Y-H (2022) Karasinger: score-free singing voice synthesis with VQ-VAE using mel-spectrograms. ICASSP, Singapore
23. Zhao W, Lu D, Lv XM, Qin L (2020) A simple data enhancement technique for singing voice synthesis. J Commun Univ China (Sci Technol) (in Chinese)
24. Zhang Z, Zheng Y, Li X, Li L (2022) Wesinger: data-augmented singing voice synthesis with auxiliary losses. INTERSPEECH, Incheon
25. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser L, Polosukhin I (2017) Attention is all you need. NeurIPS, Long Beach
26. Guo S, Shi J, Qian T, Watanabe S, Jin Q (2022) Singaug: data augmentation for singing voice synthesis with cycle-consistent training strategy. INTERSPEECH, Incheon
27. Morise M, Yokomori F, Ozawa K (2016) WORLD: a vocoder-based high-quality speech synthesis system for real-time applications. IEICE Trans Inf Syst
28. Gu Y, Yin X, Rao Y, Wan Y, Tang B, Zhang Y, Chen J, Wang Y, Ma Z (2021) Bytesing: a Chinese singing voice synthesis system using duration allocated encoder-decoder acoustic models and WaveRNN vocoders. ISCSLP (online)
29. Blaauw M, Bonada J (2020) Sequence-to-sequence singing synthesis using the feed-forward transformer. ICASSP, Barcelona
30. Shi J, Guo S, Huo N, Zhang Y, Jin Q (2021) Sequence-to-sequence singing voice synthesis with perceptual entropy loss. ICASSP, Toronto
31. Yi Ren X, Tan TQ, Luan J, Zhao Z, Liu T-Y (2020) Deepsinger: singing voice synthesis with data mined from the web. KDD, Santiago
32. Hono Y, Hashimoto K, Oura K, Nankaku Y, Tokuda K (2019) Singing voice synthesis based on generative adversarial networks. ICASSP, Brighton
33. Lee G-H, Kim T-W, Bae H, Lee M-J, Kim Y-I, Cho H-Y (2021) N-singer: a non-autoregressive Korean singing voice synthesis system for pronunciation enhancement. INTERSPEECH, Brno
34. Chandna P, Blaauw M, Bonada J, Gómez E (2019) Wgansing: a multi-voice singing voice synthesizer based on the Wasserstein-GAN. EUSIPCO, Coruna
35. Choi S, Kim W, Park S, Yong S, Nam J (2020) Korean singing voice synthesis based on auto-regressive boundary equilibrium GAN. ICASSP, Barcelona
36. Zhang Z, Zheng Y, Li X, Lu L (2022) WeSinger 2: fully parallel singing voice synthesis via multi-singer conditional adversarial training. CoRR
37. Bak T, Lee J, Bae H, Yang J, Bae J-S, Joo Y-S (2022) Generative adversarial network for artifact-free vocoder. CoRR, Avocado
38. Xue H, Wang X, Zhang Y, Xie L, Zhu P, Bi M (2022) Learn2sing 2.0: diffusion and mutual information-based target speaker SVS by learning from singing teacher. INTERSPEECH, Incheon
39. Zhang Y, Cong J, Xue H, Xie L, Zhu P, Bi M (2022) Visinger: variational inference with adversarial learning for end-to-end singing voice synthesis. ICASSP, Singapore
40. Zhang Y, Xue H, Li H, Xie L, Guo T, Zhang R, Gong C (2022) ViSinger 2: high-fidelity end-to-end singing voice synthesis enhanced by digital signal processing synthesizer. CoRR
41. Kalchbrenner N, Elsen E, Simonyan K, Noury S, Casagrande N, Lockhart E, Stimberg F, van den Oord A, Dieleman S, Kavukcuoglu K (2018) Efficient neural audio synthesis. ICML, Stockholm
42. Wang Y, Skerry-Ryan RJ, Stanton D, Wu Y, Weiss RJ, Jaitly N, Yang Z, Xiao Y, Chen Z, Bengio S, Le QV, Agiomyrgiannakis Y, Clark R, Saurous RA (2017) Tacotron: towards end-to-end speech synthesis. INTERSPEECH, Stockholm
43. Shen J, Pang R, Weiss RJ, Schuster M, Jaitly N, Yang Z, Chen Z, Zhang Y, Wang Y, Ryan R-S, Saurous RA, Agiomyrgiannakis Y, Wu Y (2018) Natural TTS synthesis by conditioning WaveNet on MEL spectrogram predictions. ICASSP, Calgary

44. Valin J-M, Skoglund J (2019) LPCNET: improving neural speech synthesis through linear prediction. ICASSP, Brighton
45. Ren Y, Yangjun Ruan X, Tan TQ, Zhao S, Zhao Z, Liu T-Y (2019) FastSpeech: fast, robust and controllable text to speech. NeurIPS, Vancouver
46. Griffin DW, Lim JS (1983) Signal estimation from modified short-time Fourier transform. ICASSP, Boston
47. Kong J, Kim J, Bae J (2020) Hifi-GAN: generative adversarial networks for efficient and high fidelity speech synthesis. NeurIPS (online)
48. Kumar K, Kumar R, de Boissiere T, Geste L, Teoh WZ, Sotelo J, de Brébisson A, Bengio Y, Courville AC (2019) MelGAN: generative adversarial networks for conditional waveform synthesis. NeurIPS, Vancouver
49. Yamamoto R, Song E, Kim J-M (2020) Parallel WaveGAN: a fast waveform generation model based on generative adversarial networks with multi-resolution spectrogram. ICASSP, Barcelona
50. Cho Y-P, Tsao Y, Wang H-M, Liu Y-W (2022) Mandarin singing voice synthesis with denoising diffusion probabilistic Wasserstein GAN. CoRR